



Verification Report

Report No : WTH25H10267886C
Applicant : Zhiwei Robotics Corp.
Address : Unit 01, 26/F, A1 Shanghai T & I Tower, No. 1699 Zhongke Road, Pudong District, Shanghai, P.R.China
Sample Name : Single Board Computer
Sample Model : LattePanda Iota
Trademark : LattePanda
Test Requested : Verify the Pb, Cd, Hg, Cr(VI), PBBs, PBDEs, DBP, BBP, DEHP, DIBP content in the sample with reference to EU RoHS Directive 2011/65/EU and its amendment Directive EU 2015/863.
Test Conclusion : PASS
Date of Receipt sample : 2025-10-13
Testing period : 2025-10-13 ~ 2025-11-08
Date of Issue : 2025-11-13
Test Result : Refer to next page (s)

Prepared By:

Waltek Testing Group HCT (Shenzhen) Co., Ltd.

Address: Building B, Tianji Industrial Park, Floor 1&2&3 No.30-9 Laiyin Road, Xinsheng Community, Longgang Street, Longgang District, Shenzhen, Guangdong, China
Tel: +86-755-84616666/400-0066-989 E-mail: service@hct-test.com

Signed for and on behalf of
Waltek Testing Group HCT (Shenzhen) Co., Ltd.

Worf Wang

Worf Wang
Waltek Testing Group HCT (Shenzhen) Co., Ltd.
<http://www.hct-test.com>

Test Result(s):
Pb, Cd, Hg, Cr(VI), PBBs, PBDEs, DBP, BBP, DEHP, DIBP content(Screen & chemical test) Test

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
1-1	Copper colored metal block	Pb	IN	31664 ^{h1}	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-2	Copper colored metal screw	Pb	IN	30350 ^{h1}	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-3	Bright silver metal screw	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	IN	ND		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-4	Bright silver metal large screw	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	IN	ND		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-5	Silver metal screw	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-6	Black printed yellow plastic sheet	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-7	Black soft rubber cable leather	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND	2025-11-06	
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-8	Red soft rubber cable leather	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		DBP	NA	ND	2025-11-06	
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-9	Silver metal wire core	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-10	White plastic	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND	2025-11-06	
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-11	Silver metal pin	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-12	White printed brown FPC	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-13	Black reinforcement sheet	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND	2025-11-06	
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-14	Silver metal shaft (Small PCB)	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-15	Brown Large Body (Small PCB)	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	IN	ND		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-16	Brown patch capacitor (Small PCB)	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-17	Black six legged body (Small PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-18	Brown printed black plastic (Small PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-19	Silver metal pins (small PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-20	Beige copper-clad laminated board with	Pb	BL	---	2025-10-13	•
		Cd	BL	---		

Report No.: WTH25H10267886C

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
	black coating (Small PCB)	Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	IN	PBBs:ND PBDEs:ND		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-21	Black adhesive tape	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-22	Grey soft rubber pad	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-23	White printed black plastic sheet	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
1-24	Black plastic shell	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	IN	PBBs:ND PBDEs:ND		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-25	Black magnetic soft adhesive	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	IN	ND		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-26	Black plastic shell	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-27	Black soft rubber cable leather	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND	2025-11-06	
		BBP	NA	ND		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		DEHP	NA	ND		
		DIBP	NA	ND		
1-28	Red soft rubber cable leather	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND	2025-11-06	
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-29	Yellow soft rubber cable leather	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND	2025-11-06	
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-30	Blue soft rubber cable leather	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-31	Silver crystal oscillator (large PCB)	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-32	Black diode (large PCB)	Pb	IN	27156 ^{h1}	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-33	Black three legged body (large PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-34	Brown patch capacitor (large PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-35	Black printed gray body (large PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-36	White printed black plastic shell (large PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	IN	PBBs:ND PBDEs:ND		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-37	Grey plastic shell (large PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-38	White plastic shell (large PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		
1-39	Silver metal casing (large PCB)	Pb	BL	---	2025-10-13	•
		Cd	BL	---		
		Hg	BL	---		

No.	Sample Description	Test Item(s)	XRF Result(s) (1)	Chemical Result(s) (2)	Received Sample(s) Date	Note
		Cr(Cr(VI))	IN	ND		
		Br(PBBs&PBDEs)	NA	---		
		DBP	NA	---		
		BBP	NA	---		
		DEHP	NA	---		
		DIBP	NA	---		
1-40	Blue plastic shell (large PCB)	Pb	BL	---	2025-10-13	●
		Cd	BL	---		
		Hg	BL	---		
		Cr(Cr(VI))	BL	---		
		Br(PBBs&PBDEs)	BL	---		
		DBP	NA	ND		
		BBP	NA	ND		
		DEHP	NA	ND		
		DIBP	NA	ND		

Note:

●=Actual tested sample

As specified by client, only test the designated sample.

^{h1}= According to the declaration from the client, Lead(Pb) in specimen(s) is(are) exempted by EU RoHS Directive 2011/65/EU and its amendment Directive EU 2015/863 base on:6(c):Copper alloy containing up to 4 % lead by weight.

Remark:

(1): ED-XRF test

(a) For the restricted substances PBBs/PBDEs, the ED-XRF results show the total Br content; for the restricted substance Cr(VI), the ED-XRF results show the total Cr content.

(b) Results were obtained by ED-XRF for primary screening, and further chemical testing are recommended to be performed, if the concentration exceeds the below warning value according to IEC 62321-3-1:2013.

Unit: mg/kg

Elements	Polymers	Metals	Composite material
Pb	BL ≤ 100 < X < 1200 ≤ OL	BL ≤ 100 < X < 1200 ≤ OL	BL ≤ 80 < X < 1300 ≤ OL
Cd	BL ≤ 30 < X < 120 ≤ OL	BL ≤ 30 < X < 120 ≤ OL	BL ≤ 30 < X < 120 ≤ OL
Hg	BL ≤ 100 < X < 1200 ≤ OL	BL ≤ 100 < X < 1200 ≤ OL	BL ≤ 80 < X < 1300 ≤ OL
Cr	BL ≤ 200 < X	BL ≤ 200 < X	BL ≤ 150 < X
Br	BL ≤ 200 < X	---	BL ≤ 200 < X

(c) BL=Below Limit by ED-XRF analysis, OL=Over Limit by ED-XRF analysis, IN=Inconclusive, NA=Not Applicable, --- = Not regulated, X=need further chemical analysis

(d) For composite material, the ED-XRF results may be different to the actual content in the sample.

(2): Chemical test and regulatory limits

Test Items		Test Method	Test Equipment	MDL	Limit(by weight in homogeneous materials)
Pb		IEC 62321-5:2013	ICP-OES/AAS	2mg/kg	1000mg/kg
Cd		IEC 62321-5:2013	ICP-OES/AAS	2mg/kg	100mg/kg
Hg		IEC 62321-4:2013+AMD1:2017	ICP-OES	2mg/kg	1000mg/kg
Cr(VI)	Metal	IEC 62321-7-1:2015	UV-VIS	0.10µg/cm ²	1000mg/kg
	Others	IEC 62321-7-2:2017	UV-VIS	8mg/kg	
PBBs		IEC 62321-6:2015	GC-MS	5mg/kg(Each)	1000mg/kg(Sum)
PBDEs		IEC 62321-6:2015	GC-MS	5mg/kg(Each)	1000mg/kg(Sum)
DBP		IEC 62321-8:2017	GC-MS	30mg/kg	1000mg/kg
BBP		IEC 62321-8:2017	GC-MS	30mg/kg	1000mg/kg
DEHP		IEC 62321-8:2017	GC-MS	30mg/kg	1000mg/kg
DIBP		IEC 62321-8:2017	GC-MS	30mg/kg	1000mg/kg

(a) Unit of Chemical test result: Cr(VI) of metal:µg/cm², Other:mg/kg;

µg/cm² (microgram per square centimeter)

1mg/kg=1ppm=0.0001%, mg/kg (milligram per kilogram) = ppm (parts per million)

ND=not detected (less than method detection limit),

MDL=Method Detection Limit, --- =Not Conducted

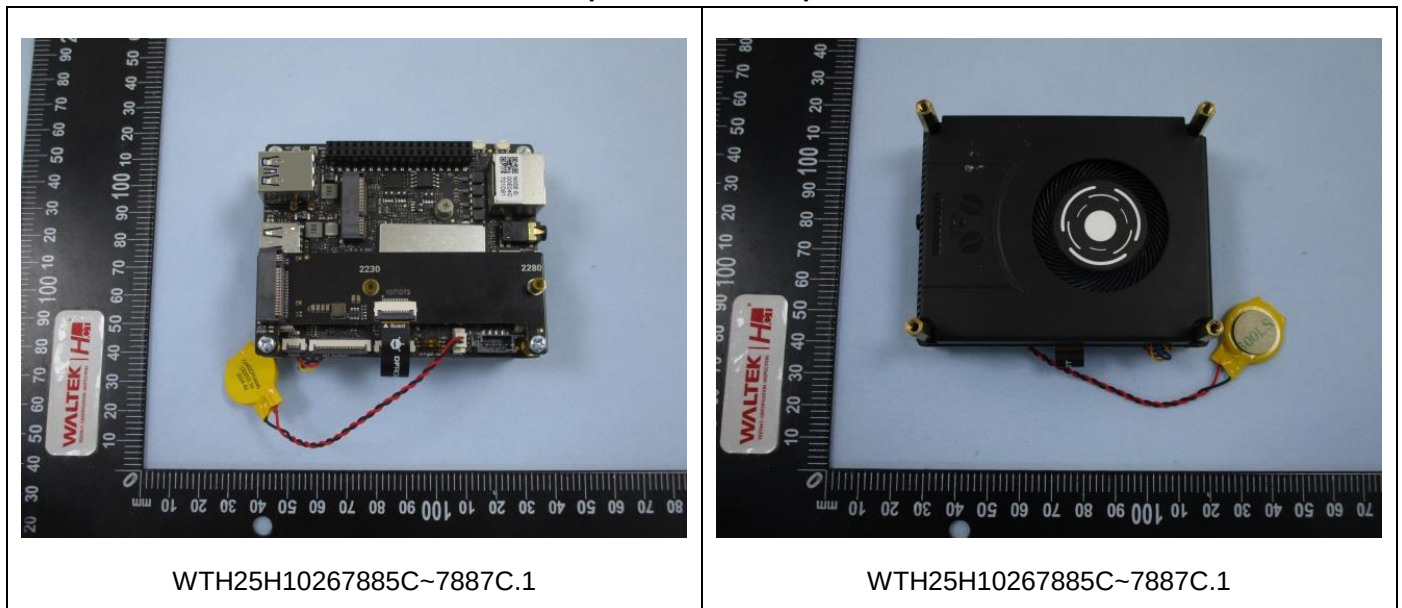
(b) For corrosion-protected coatings on metals, according to the test method IEC 62321-7-1:2015: ①The sample is positive for Cr(VI) if the Cr(VI) concentration is greater than 0.13µg/cm². The sample coating is considered to contain Cr(VI); ②The sample is negative for Cr(VI) if Cr(VI) is ND (concentration less than 0.10µg/cm²). The coating is considered a non-Cr(VI) based coating; ③The result between 0.10µg/cm² and 0.13µg/cm² is considered

Report No.: WTH25H10267886C

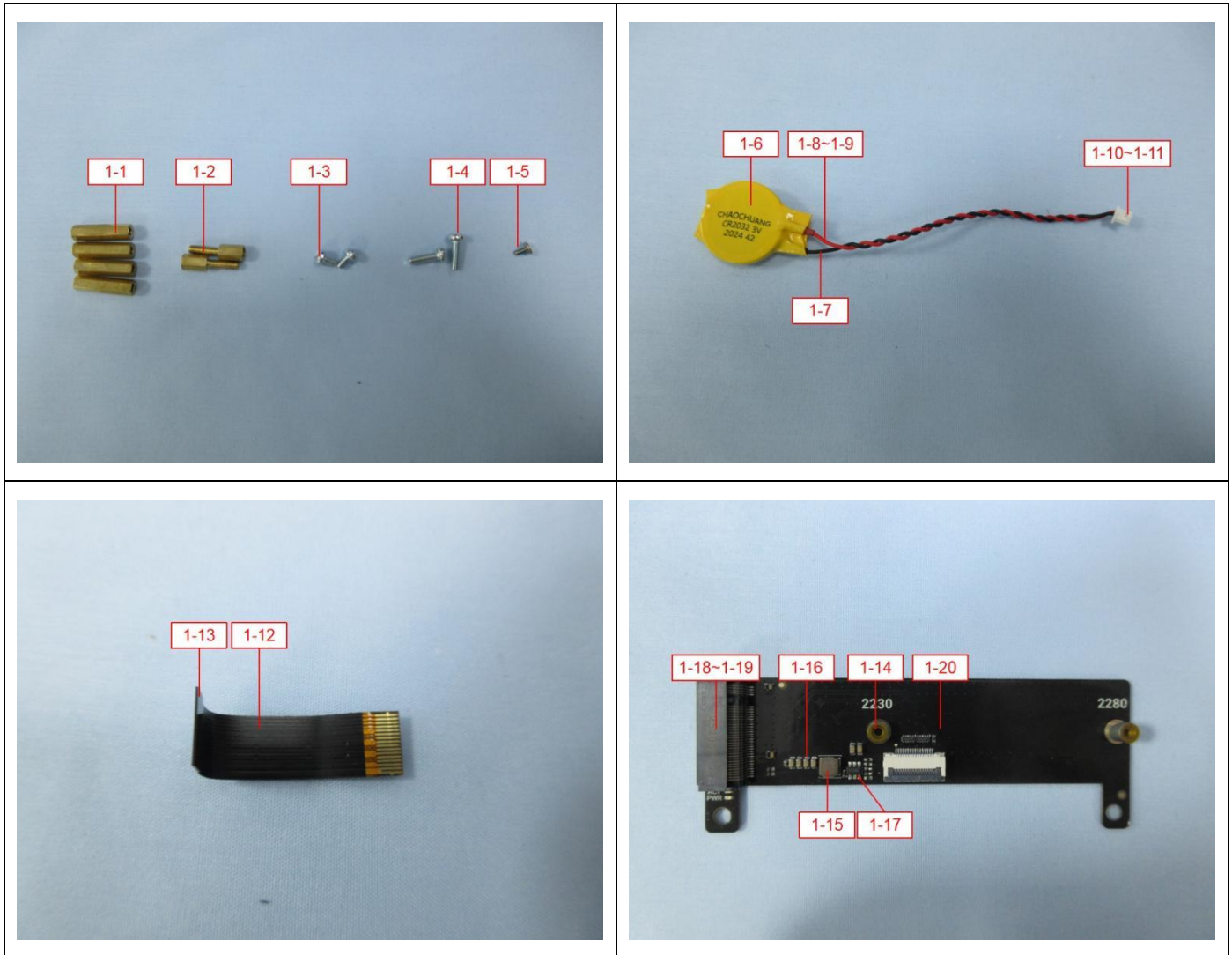
to be inconclusive -unavoidable coating variations may influence the determination; The surface difference of samples from different individuals may affect the determination results of Cr(VI); Information on storage conditions and production date of the tested sample is unavailable and thus Cr(VI) results represent status of the sample at the time of testing.

(3): Screening and chemical tests were performed for the samples indicated by the photo in this report.

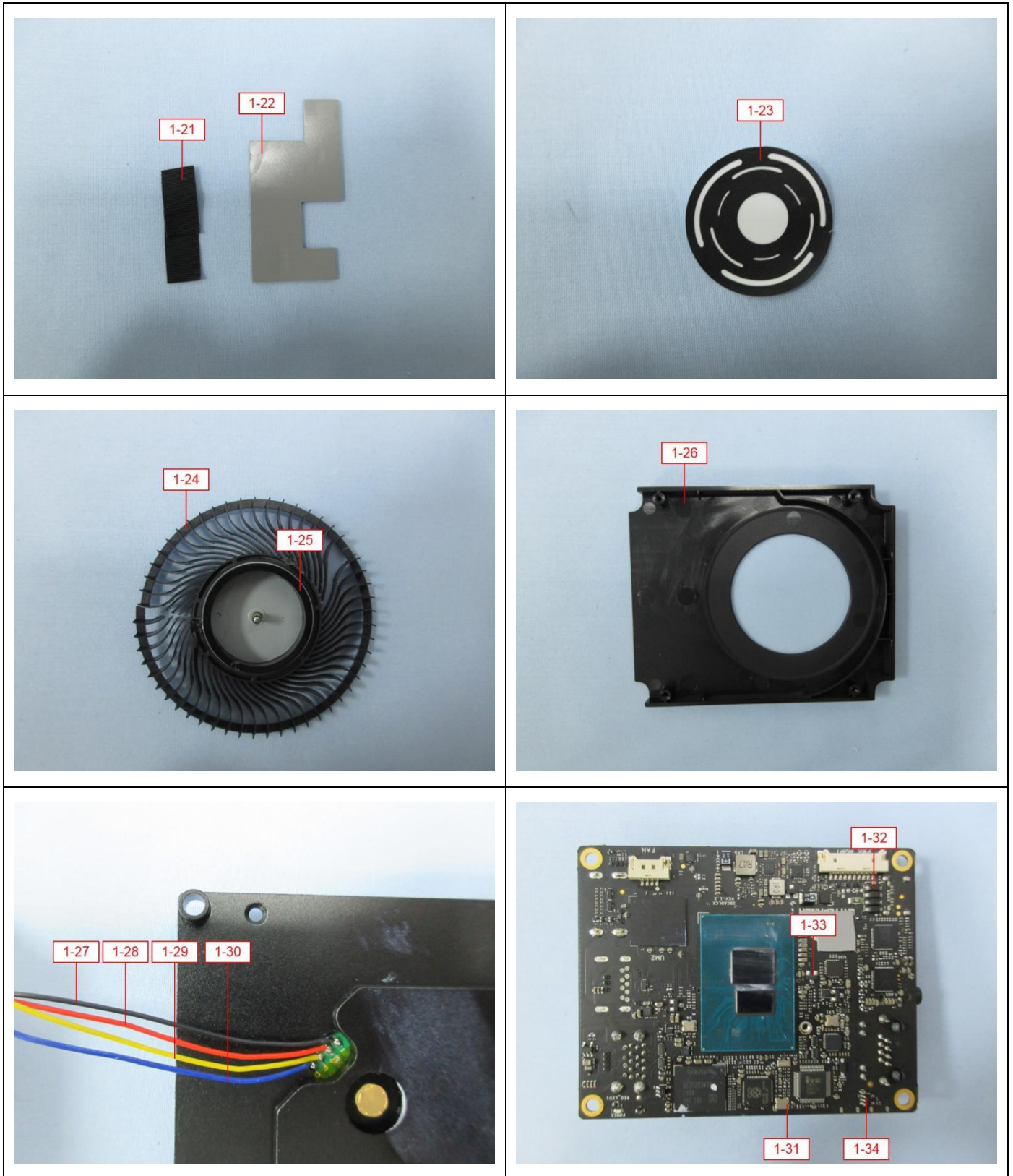
The photo of the sample



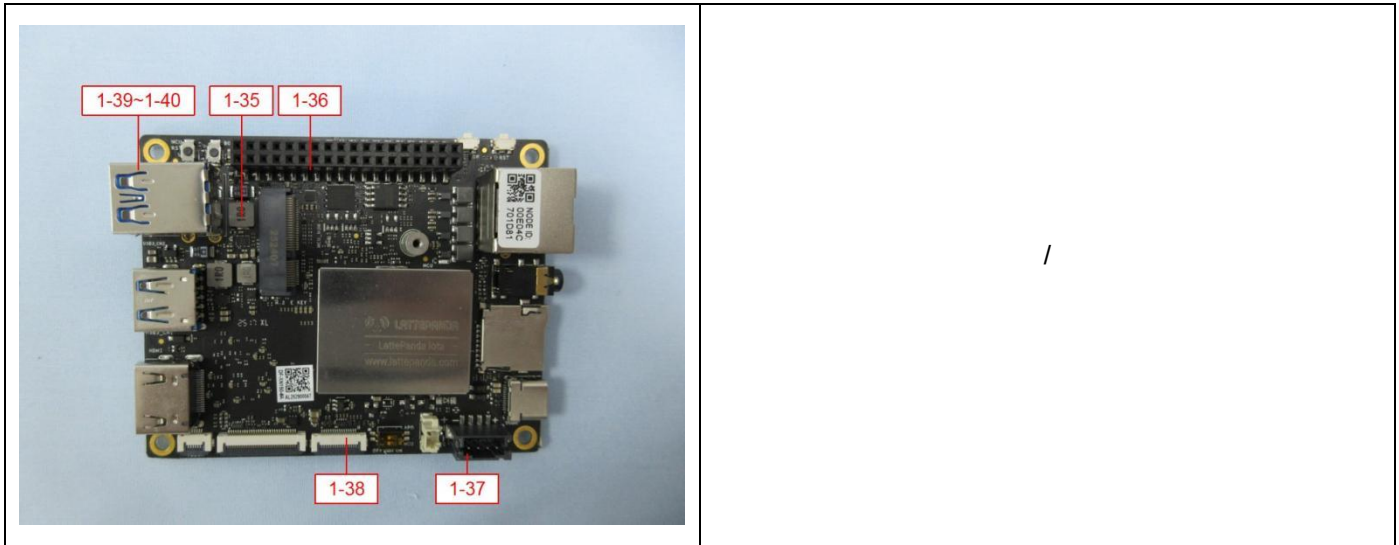
The photo of Disassembly



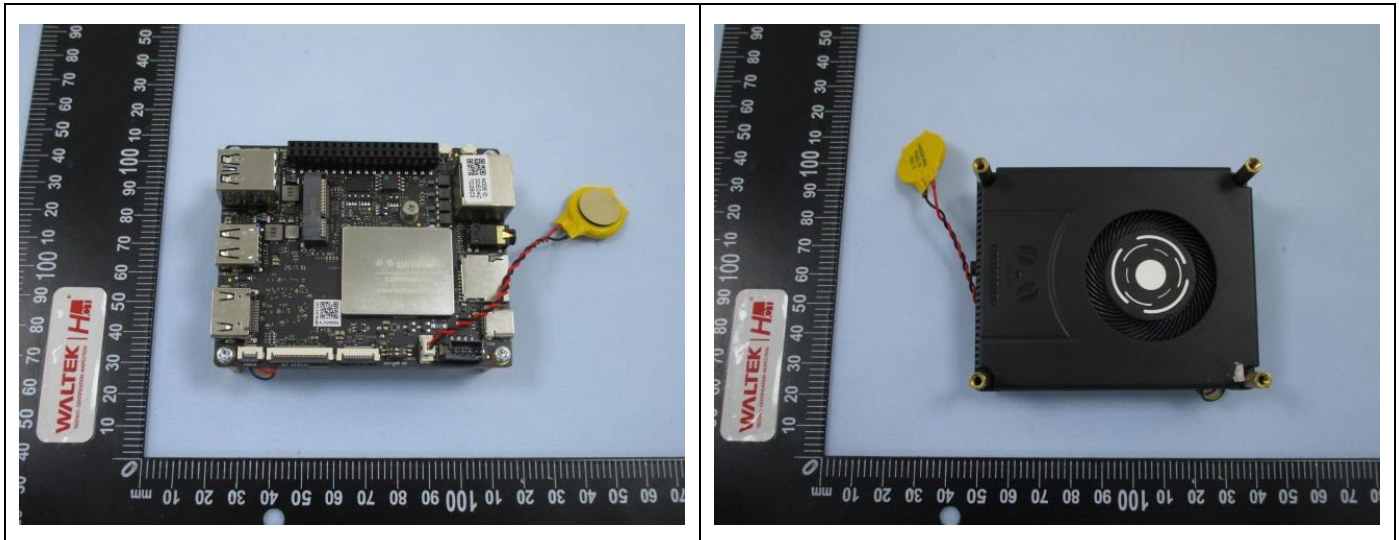
Report No.: WTH25H10267886C



Report No.: WTH25H10267886C



Client's reference photo(Non-tested sample)



Statement:

- ❖ This report is considered invalid without approved signature and special seal.
- ❖ The Applicant name and Address, the sample(s) and sample information was/were provided by the applicant who should be responsible for the authenticity which HCT hasn't verified.
- ❖ The result(s)(conclusion) shown in this report refer(s) only to the sample(s) tested.
- ❖ Without written approval of HCT, this report can't be reproduced except in full.
- ❖ Test results in report are intended for purposes such as scientific research, teaching, internal quality control, product research and development, etc...

Report No.: WTH25H10267886C

➤ Decision rules used in this report:

- (1) According to the Decision rules in the regulations/standards listed in the Test Requested;
- (2) If there is no Decision rules specified in the regulations listed in the Test Requested, then according to CNAS-GL015 Guidelines on Decision Rules and Statements of Conformity, 6.2.1, Simple Acceptance ($w=0$) of The binary Decision rule:
PASS (Accepted) - The measured value is within the tolerance interval.
FAIL (Rejected) - The measured value is outside the tolerance interval.

===== End of Report =====